FORKING INDEPENDENCE COINCIDES WITH TENSOR INDEPENDENCE IN THE R-SPACE THEORIES OF THE PAPER

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Editorial note. Extended pairing, per the Chief's ruling of 2026-10-12: the author's proof environment ends by announcing a claim, so the claim environment (source lines 1124-1128) and the proof that follows it (1133-1169) are counted as the continuation of the same proof and are carried as additional proof spans; the sentence closing the main proof is kept verbatim.

Editorial note. The only reference to an outside item inside the counted spans is Lemma aclopen (proof line 1153), recorded as a navigation stub of kind lemma; the line that closes the proof by referring back to this lemma's own label lies outside the counted ranges.

Allowed standard prerequisites (editorial).

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- The two independence notions compared: forking independence over C and the tensor independence defined by free amalgamation of algebraic closures. Set up in the discussion preceding the counted statement, ending at source line 1087. Fixes both sides of the equivalence and the notation used throughout the proof.
- The r-space theories of the paper and their algebraically closed sets. Used inside the counted proof at lines 1133-1169. Supplies the behaviour of algebraic closures and neighbours that the continuation of the proof uses.

Source reference. Original lemma 4.22: sources/logic-fl03-simple-homogeneous-structures-and/pinned-original-source.tex, source lines 1153–1153. Navigation pointer for \ref{aclopen} used inside the counted span (line 1153); the argument that consumes it is retained in the span.

Lemma 0.1. $A \mathop{\smile\!\!\!\!\!\!|}_C B$ if and only if $A \mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C B$.

Proof. First, if $A \mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C B$ with $C \leq M$, then $A \mathop{\smile\!\!\!\!\!\!|}_C B$ because clearly, $\mathrm{tp}(A/CB)$ extends to a C-invariant global type. Now suppose that $A \mathop{\smile\!\!\!\!\!\!|}_C B$. Then, $A \mathop{\smile\!\!\!\!\!\!|}_C \mathrm{acl}(BC)$, so we may assume $C \leq B$ and B is algebraically closed. Now, let $A' = \mathrm{acl}(AC)$ and suppose that $A \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C B$. We show that $A \not\mathop{\smile\!\!\!\!\!\!|}_C B$. We may produce a C-indiscernible sequence $\{B_i\}_{i<\omega}$ with $B_0 = B$ so that, for $i < \omega$, $B_i \mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C \{B_j\}_{j<i}$; for $p(X, B) = \mathrm{tp}(A/B)$, it suffices to show that $\cup\{p(X, B_i)\}_{i<\omega}$ is inconsistent. Suppose otherwise; then we may in fact assume that $\{B_i\}_{i<\omega}$ is indiscernible over A'. We then get a contradiction from the following claim: □

Claim 0.2. (Here we still assume $A \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C B$ with $C \leq B$, B algebraically closed.) Let $\{B_i\}_{i<\omega}$ be a $A' = \mathrm{acl}(AC)$-indiscernible sequence with $B_0 = B$. Then $B_n \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C \{B_i\}_{i<n}$.

Proof. We may assume that $B_i \cap B_j = C$ for $i \neq j < \omega$; then clearly, $A' \cap B = C$. It will suffice to show that $\mathrm{acl}(\{B_i\}_{i<n+1}) \supsetneq \cup_{i<n+1} B_i$. If $A \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C B$, either $\mathrm{acl}(A'B) = A'B$ and A' and B are not freely amalgamated over C, or $\mathrm{acl}(A'B) \supsetneq A'B$. In the first case, there is some $a \in A' \backslash C$ with a neighbor b in $B \backslash C$. Then $\{a\}$ will have $n+1$ neighbors in $\{B_i\}_{i<n+1}$, so will belong to $\mathrm{acl}(\{B_i\}_{i<n+1})$ but will not belong to $\cup_{i<n+1} B_i$. Therefore, $B_n \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C \{B_i\}_{i<n}$. In the second case, since $\{B_i\}_{i<\omega}$ and thus $\{\mathrm{acl}(A'B_i)\}_{i<\omega}$ is indiscernible over A', there is some fixed $D \supseteq A'$ so that $\mathrm{acl}(A'B_i) \cap \mathrm{acl}(A'B_j) = D$ for $i \neq j \leq \omega$, and so that $\{B_i\}_{i<\omega}$ is indiscernible over D. As we are in the second case, there are also $\{F_i\}_{i<\omega}$, so that $F_i \subseteq \mathrm{acl}(A'B_i) \backslash A'B_i$ is finite and closed over $A'B_i$ for $i < \omega$, and so that $\mathrm{acl}(A'B_i)F_i \equiv \mathrm{acl}(A'B_j)F_j$ for $i \neq j < \omega$. Suppose first that for some (every) $i < \omega$, $F_i \subseteq D$. Each of the F_i must have neighbors in B_i, since F_i is closed over $A'B_i$ and A' is algebraically closed and so $A' \leq \mathbb{M}$ by Lemma 4.22. So if $F_i \subseteq D$, then D and B are not freely amalgamated over C, and $B_n \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C \{B_i\}_{i<n}$ by the first case. Otherwise, for some (every) $i < \omega$, $F_i \not\subseteq D$. There is some finite $D_0 \subseteq D$ so that, for all $i < \omega$, D_0 consists of all neighbors of $F_i \backslash D$ in D. Then $F = D_0 \cup_{i=0}^{n} (F_i \backslash D)$ is a finite set, not contained in $\cup_{i<n+1} B_i$, so that every point of F has at least $n+1$ neighbors in $F \cup (\cup_{i<n+1} B_i)$: each point of F_i has $n+1$ neighbors that are either in $D \supseteq A$ (so in $D_0 \subseteq F$), in $F_i \backslash D$, or in B_i, while each point of D_0 has a neighbor in each of the $F_i \backslash D \subseteq F$, and therefore has at least $n+1$ neighbors in F. So $\mathrm{acl}(\{B_i\}_{i<n+1}) \supsetneq \cup_{i<n+1} B_i$, because $F \subseteq \mathrm{acl}(\{B_i\}_{i<n+1})$, and again $B_n \not\mathop{\smile\!\!\!\!\!\!|}^{\otimes}_C \{B_i\}_{i<n}$.

□